

Back

Yossef Teshome

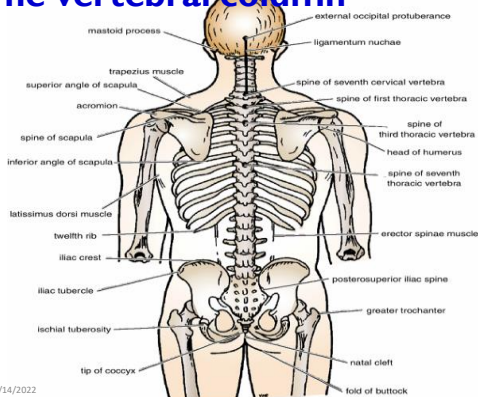
The Back

- Can be defined as the posterior surface of the trunk.
- It includes
 - ✓ Skin and subcutaneous tissue
 - ✓ **Muscles:** a superficial layer & deeper layers mm
 - ✓ **Vertebral column:** vertebrae, IVDs & associated ligaments
 - ✓ Ribs: particularly their posterior portions, **medial to the angles of the ribs.**
 - ✓ Spinal cord and meninges
 - ✓ Various segmental nerve & blood vessel
- Supported mainly by vertebral column

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The vertebral column



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The vertebral column

Function

- Protects the spinal cord and spinal nerves
- Supports the weight of the body superior to the level of the pelvis
- Provides a partly rigid and flexible axis for the body
- Provide a site on which the head is placed and pivots
- Plays an important role in posture and locomotion

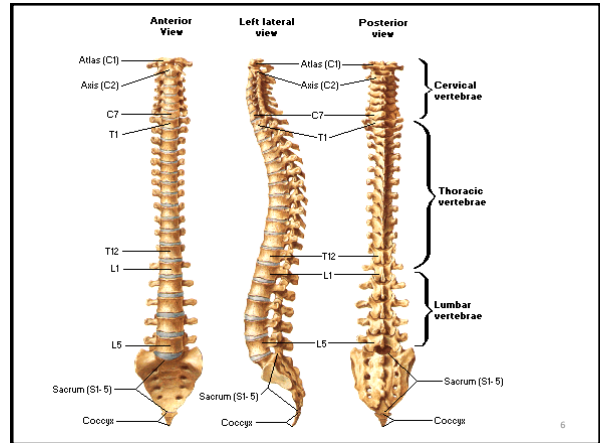
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Composition of vertebral column

- The vertebral column in an adult typically consists of **26 vertebrae**
 - become larger as the vertebral column descends to the sacrum and then become progressively smaller toward the apex of the coccyx
 - **Intervertebral discs** are pads of fibrocartilage that bind together the vertebral bodies of adjacent vertebrae.
 - Divided in to five regions:
 - ✓ Cervical7
 - ✓ Thoracic12
 - ✓ Lumbar5
 - ✓ Sacral 1 (five fused vertebrae)
 - ✓ Coccygeal..... 1 (four fused vertebrae)
- Total of 26 vertebrae

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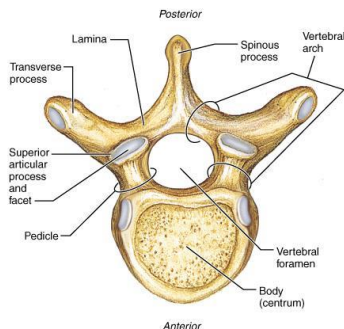


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Characteristics of Typical Vertebra

Consists of:-

- Vertebral body.**
- Vertebral arch and**
- Seven processes**



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Characteristics of Typical Vertebra

I. Vertebral body

- More massive & roughly cylindrical; weight bearing part of the vertebra
- linked to adjacent vertebral bodies by **IVD** and **ligaments**, forming the **cartilaginous joints**.
- Thoracic vertebrae has **costal facets**, which articulate with the heads of the corresponding and subjacent (just below) ribs.
- The size of vertebral bodies increases inferiorly as the amount of weight supported increases.
 - most markedly from T4 inferiorly

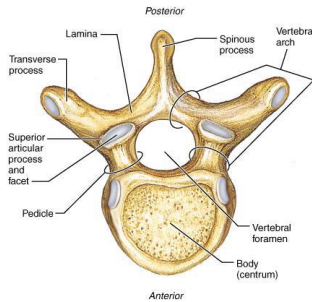
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Characteristics of Typical Vertebra

II. Vertebral arch

- is posterior to the vertebral body and
- forms the lateral and posterior parts of the **vertebral foramen**.
- consists of two (R&L) pedicles and laminae
- has seven processes



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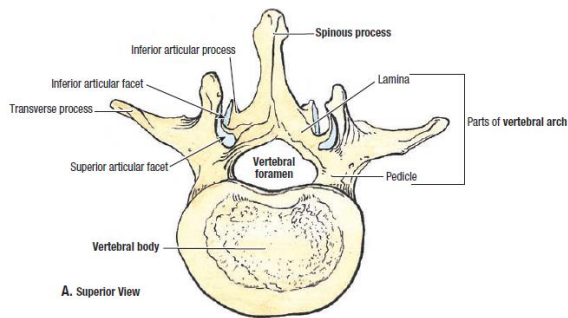
Characteristics of Typical Vertebra...

II. Vertebral arch...

- The vertebral foramina of all the vertebrae together form the **vertebral canal**, which contains
 - Spinal cord, spinal nerve roots and spinal meninges
 - Blood vessels and
 - It continuous superiorly with the cranial cavity, through the **foramen magnum** of the skull.
- Consists of pedicles and laminae:
 - two **pedicles**; bony pillars that attach the vertebral arch to the vertebral body;
 - two **laminae**; flat sheets of bone that extend from each pedicle to meet in the midline and form the roof of the vertebral arch.

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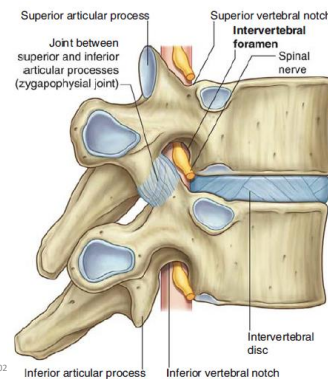
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Characteristics of Typical Vertebra



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Characteristics of Typical Vertebra

Seven processes:

- I. **One median spinous process**
 - ✓ Projects posteroinferiorly from the vertebral arch at the junction of the laminae
 - ✓ It typically overlapping the vertebra below
 - II. **Two transverse processes project posterolaterally from the junctions of the pedicles and laminae**
- ❖ The spinous and transverse processes,
- ✓ afford attachments for deep back muscles and serve as levers

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Characteristics of Typical Vertebra

Seven processes...

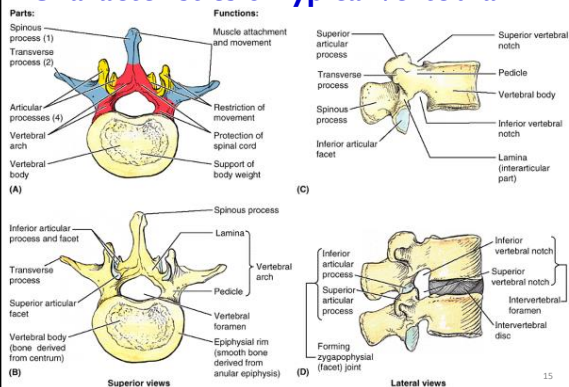
III. Four articular processes (G. zygapophyses)

- ✓ two superior and two inferior arise from the junctions of the pedicles and laminae
 - ✓ each bearing an articular surface (facet)
- ❖ **Vertebral notches (sup. & inf.)** are indentations on each pedicle superiorly and inferiorly
- ✓ b/n the superior and inferior articular processes posteriorly and the corresponding projections of the body anteriorly

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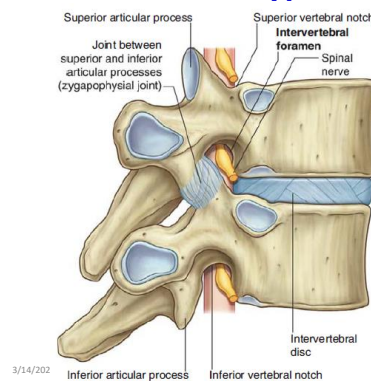
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Characteristics of Typical Vertebra



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Characteristics of Typical Vertebra



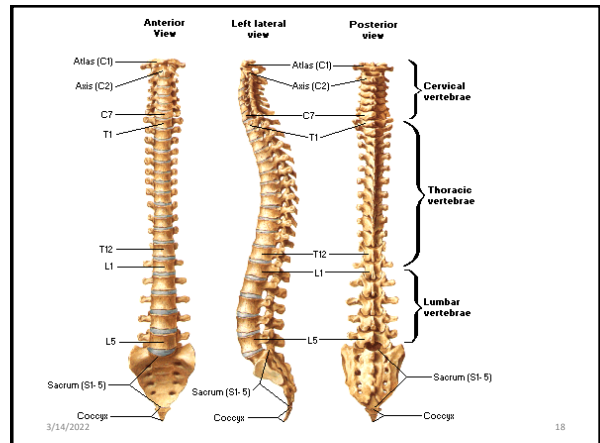
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Regional Characteristics of the Vertebrae

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Cervical Vertebrae

- The smallest of the 24 movable vertebrae
- Their smaller size reflects the fact that they bear less weight than do the larger inferior vertebrae
- Although the cervical IV discs are thinner than those of inferior regions, they are relatively thick compared to the size of the vertebral bodies they connect
- Seven cervical vertebrae form the cervical region of the vertebral column & classified as
 - Typical (3rd -6th) and
 - Atypical (1st, 2nd & 7th)

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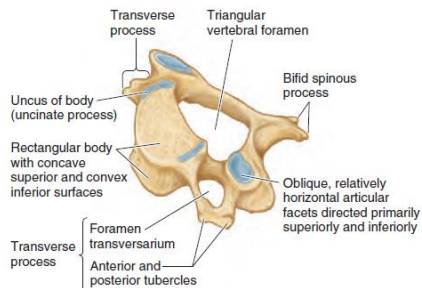
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X-features of Typical Cervical Vertebrae

- **Transverse processes** possess a foramen transversarium for the passage of the vertebral artery and veins
 - note that the vertebral artery passes through the transverse processes C1 to 6 and not through C7
 - ✓ It ends laterally in two projections: anterior & posterior tubercles.
 - ✓ The anterior tubercles of C6 are called **carotid tubercles**
 - **b/se** the common carotid arteries may be compressed here to control bleeding from these vessels.
- The **spines** are small (short) and bifid
- The **body** is small and broad from side to side
 - has a concave superior surface & a convex inferior surface
- The **vertebral foramen** is large and triangular
- **Articular processes**
 - ◆ **Superior facets** - face backward and upward;
 - ◆ **Inferior facets** - face downward and forward.

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X-Cs of Typical Cervical Vertebrae



(D) Superior oblique view of typical cervical vertebra

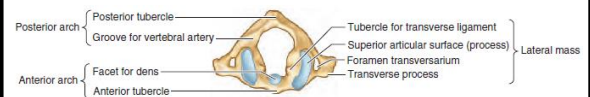
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X-Cs of the Atypical Cervical Vertebrae

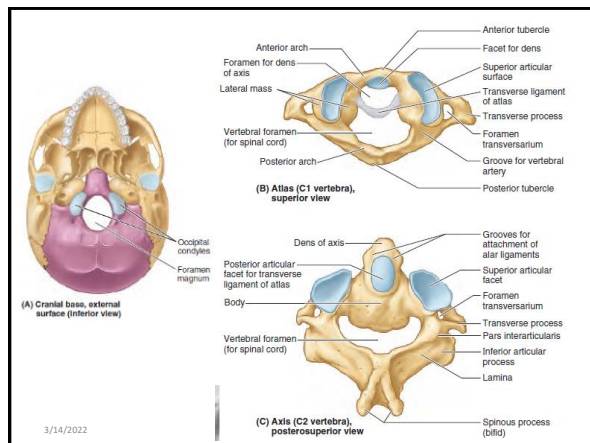
C1 or Atlas:

- Does not possess a **body** or a **spinous process**
- Consisting of **two lateral masses** connected by **anterior & posterior arches**.
- The lateral mass has
 - Concave superior articular facets articulate with the occipital condyles (**atlanto-occipital joints**) and
 - Inferior articular facets articulate with the axis (**atlantoaxial J**)



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X-Cs of the Atypical Cervical Vertebrae

C2 or axis:

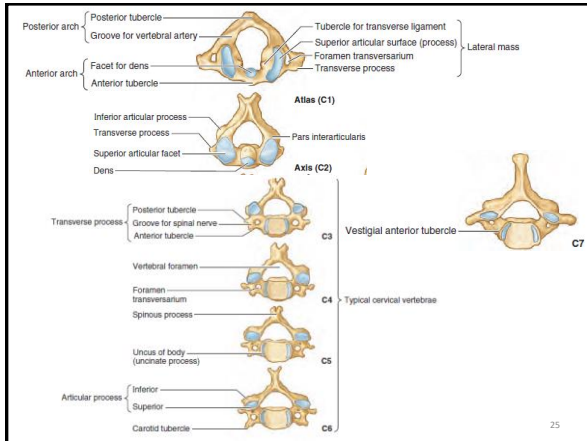
- Has a peg like dens (odontoid) process that projects from the superior surface of the body
 - representing the body of the atlas that has fused with the body of the axis

C7 or vertebra prominens:

- Is so named because it has the **longest spinous process**, that is nearly horizontal forms a visible protrusion and it is **not bifid**
- The transverse process is large, but the **foramen transversarium** is small and transmits **vertebral vein**
- Provides an attachment site for the **ligamentum nuchae**, **supraspinous ligaments**, and **numerous back muscles**.

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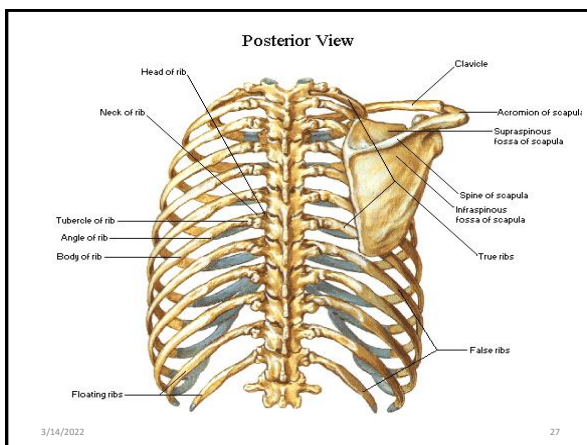


Thoracic Vertebrae

- The thoracic vertebrae lie in the upper back and provide attachment for the ribs
- There are twelve thoracic vertebrae (T1-T12), each of which is characterized by articulations with ribs

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Thoracic Vertebrae

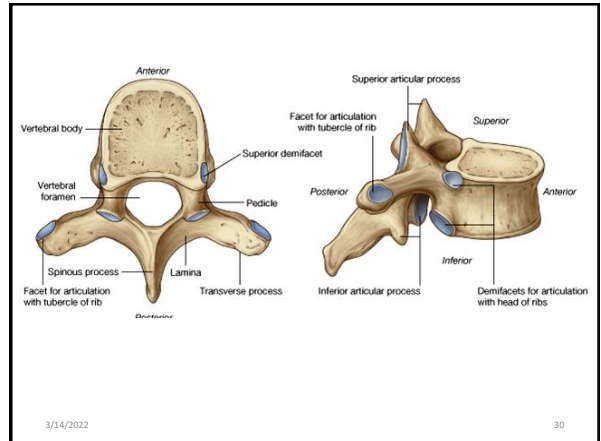
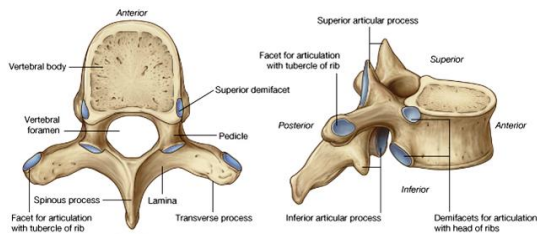
- The middle four thoracic vertebrae (T5 - T8) demonstrate all the features typical of thoracic vertebrae
 - **Vertebral body**
 - ✓ Heart shaped; **costal facets** for articulation with head of rib
 - **Vertebral foramen**
 - ✓ Circular and smaller than those of cervical and lumbar vertebrae
 - **Transverse processes**
 - ✓ Long and strong and extend posterolaterally;
 - ✓ length diminishes from T1 to T12
 - ✓ **T1-T10 have facets** for articulation with tubercle of rib

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Thoracic Vertebrae

- **Articular processes:** Nearly vertical articular facets;
 - **Superior facets** directed posteriorly and slightly laterally;
 - **Inferior facets** directed anteriorly and slightly medially;
- **Spinous process:** Long; slope postero-inferiorly; tips extend to level of vertebral body below



Thoracic Vertebrae

- The T1 - T4 vertebrae share some features of cervical vertebrae
 - T1 is atypical of thoracic vertebrae in that it has a long, almost horizontal spinous process that may be nearly as prominent as that of the vertebra prominens (C7)
 - **Body of T1** has
 - ✓ Complete superior costal facet for the 1st rib and
 - ✓ Inferior demifacet that contributes to the articular surface for the 2nd rib

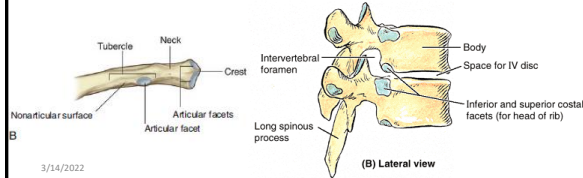
Thoracic Vertebrae

- The T9–T12 vertebrae have some features of lumbar vertebrae
 - e.g. → Tubercles similar to the accessory processes
 - Mammillary processes (small tubercles) also occur on vertebra T12.
- Most of the transition in characteristics of vertebrae from the thoracic to the lumbar region occurs over the length of a single vertebra: vertebra T12.
 - Its superior half is thoracic in character,
 - having costal facets and articular processes that permit primarily rotatory movement, whereas
 - Its inferior half is lumbar in character,
 - devoid of costal facets and having articular processes that permit only flexion and extension.
- Consequently, vertebra T12 is subject to transitional stresses that cause it to be the most commonly fractured vertebra.

Thoracic Vertebrae

Articulation with ribs

- A typical thoracic vertebra has three sites on each side for articulation with ribs:
 - Two demifacets** (i.e. partial facets)
 - located on the superior and inferior aspects of the body
 - articulation sites for the heads of adjacent ribs



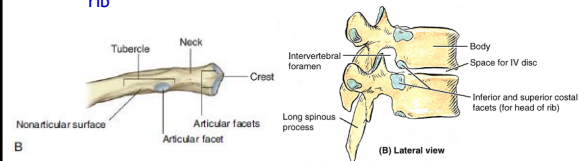
Thoracic Vertebrae

Articulation with ribs

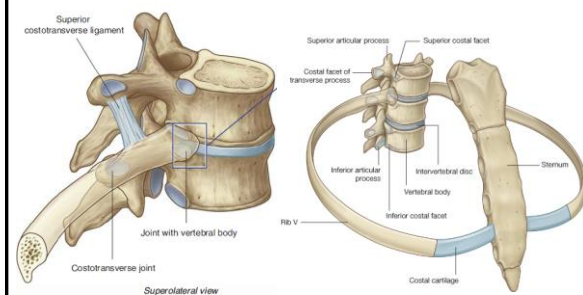
I. Two demifacets...

- ✓ **Superior costal facet** articulates with part of the head of its own rib, &
- ✓ **Inferior costal facet** articulates with part of the head of the rib below

- An oval facet (**transverse costal facet**) at the end of the transverse process articulates with the **tubercle of its own rib**



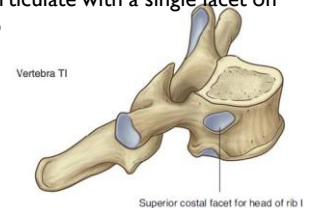
Thoracic Vertebrae



Thoracic Vertebrae

Articulation with ribs

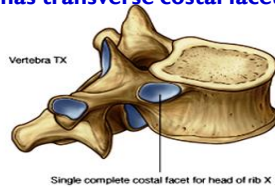
- ❖ Not all vertebrae articulate with ribs in the same fashion
- The superior costal facets on the body of vertebra T1 are complete and articulate with a single facet on the head of its own rib
- In other words, the head of Rib I does not articulate with C7 vertebra



Thoracic Vertebrae

Articulation with ribs

- T10 vertebra (and often T9) articulates only with its own ribs and therefore lacks inferior demifacets on the body
- **But has transverse costal facets**



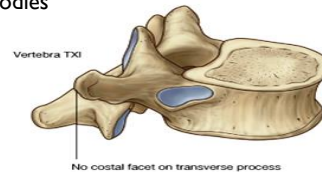
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Thoracic Vertebrae

Articulation with ribs

- T11 and T12 vertebrae articulate only with the heads of their own ribs
- **They lack transverse costal facets**
- Have only a single complete facet on each side of their bodies



No costal facet on transverse process

Lumbar Vertebrae

- Lumbar vertebrae are in the lower back between the thorax and sacrum
- The vertebral bodies are massive and progressively increase in size from vertebra L1 to L5
- *Note that the lumbar vertebrae have no facets for articulation with ribs and no foramina in the transverse processes*
 - **Vertebral body**
 - ✓ Massive; kidney shaped when viewed superiorly
 - **Vertebral foramen**
 - ✓ Triangular; larger than in thoracic vertebrae and smaller than in cervical vertebrae

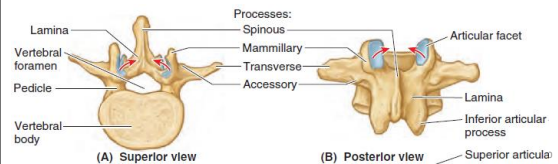
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Lumbar Vertebrae

➤ Articular processes

- ✓ Nearly vertical facets;
 - **Superior facets** directed posteromedially (or medially);
 - **has mammillary process**; give attachment to both multifidus & intertransversarii muscles of the back.
 - **Inferior facets** directed anterolaterally (or laterally);



Lumbar Vertebrae

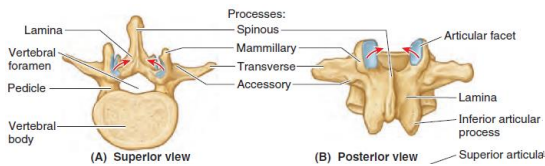
➤ Transverse processes

✓ Long and slender;

✓ **Accessory process** on posterior surface of base of each process; provides an attachment for the **intertransversarii muscles**.

➤ Spinous process

✓ Short, flat, and quadrangular and project backward



Sacrum

- A large, triangular, wedge-shaped bone
- Composed of five fused sacral vertebrae in adults.
- Is located between the hip bones and forms the roof and posterosuperior wall of the posterior pelvic cavity
 - provides strength and stability to the pelvis.
 - transmits the weight of the body to the pelvic girdle,
- The sacrum is often **wider in proportion to length** in female than in male, but the body of the S1 vertebra is usually larger in males.

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Sacrum

Base of the sacrum

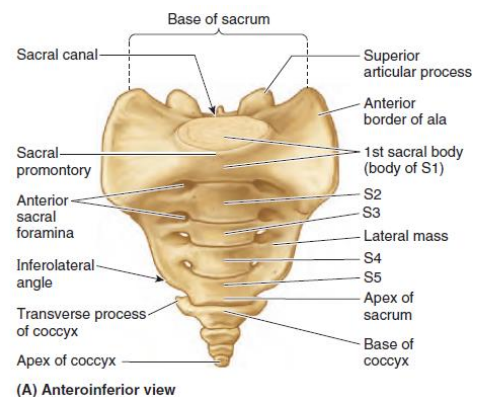
- Formed by the superior surface of the S1 vertebra.
- Its superior articular processes articulate with the inferior articular processes of the L5 vertebra.
- **Sacral promontory** (L. mountain ridge),
 - the prominent anterior edge of the S1 vertebra, an important **obstetrical landmark**.

Apex of the sacrum

- tapering inferior end, has an oval facet for articulation with the coccyx

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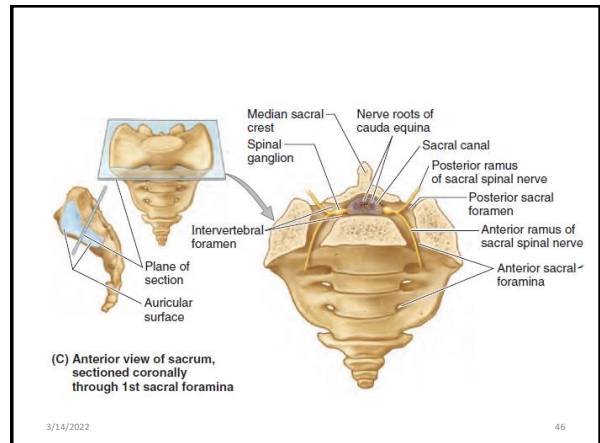
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Sacrum

- Has four pairs of **sacral foramina** for the exit of the ventral and dorsal primary rami of the first four sacral nerves.
 - The anterior (pelvic) sacral foramina are larger than the posterior (dorsal) ones.
- Sacral canal**
 - Is the continuation of the vertebral canal
 - Contains the bundle of spinal nerve roots arising inferior to the L1 vertebra, known as the **cauda equina** (L. horse tail), that descend past the termination of the spinal cord

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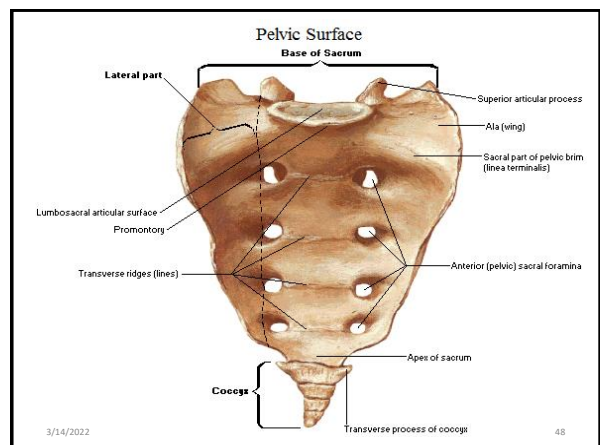
Sacrum

Pelvic surface of the sacrum

- It is smooth and concave
- During childhood, the individual sacral vertebrae are connected by **hyaline cartilage** and separated by **IV discs**
- In adults there is **four transverse lines** indicating where fusion of the sacral vertebrae occurred
- Fusion starts after age 20;
 - However, most of the IV discs remain **unossified up to or beyond middle life**

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Sacrum

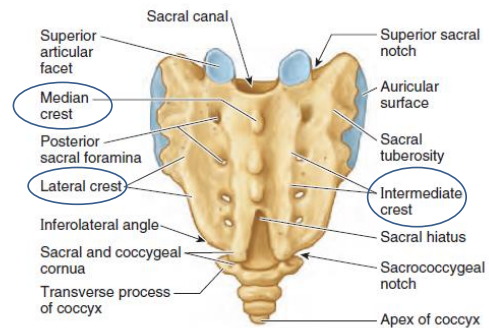
Dorsal surface of the sacrum

- It is rough, convex, and marked by five prominent longitudinal ridges.
 1. **Median sacral crest** (central ridge),
 - represents the fused rudimentary spinous processes of the superior three or four sacral vertebra; S5 has no spinous process.
 2. **Intermediate sacral crests**
 - represent the fused articular processes, and
 3. **lateral sacral crests**
 - are the tips of the transverse processes of the fused sacral vertebrae.

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Dorsal surface of the sacrum



(B) Posterosuperior view

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Sacrum

- The clinically important features of the dorsal surface of the sacrum are the inverted U-shaped **sacral hiatus** and the **sacral cornua** (L. horns).

Sacral hiatus:

- Results from the absence of the laminae and spinous process of S5 and sometimes S4.
- It is used for the administration of **caudal (extradural) anesthesia**.
- **Posterior sacral foramina** also used for **trans sacral epidural anesthesia**.

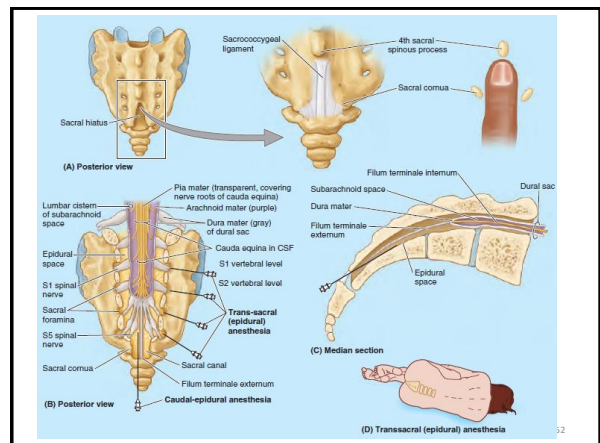
→ Epidural anesthesia may be applied during childbirth

Sacral cornua or horn:

- Representing the inferior articular processes of S5 vertebra
- It is an important landmark for locating the sacral hiatus.

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Coccyx (tail bone)

- A small triangular bone formed by fusion of the four rudimentary coccygeal vertebrae.
- It is remnant of **caudal eminence** of embryonic tail-like skeleton,
 - present in embryos from the end of the 4th wk until the beginning of 8th wk.
- Provides attachment for parts of the gluteus maximus and anococcygeal ligament, coccygeus and levator ani muscles.
- Does not support body weight when standing; however, when sitting it receives some weight

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Coccyx

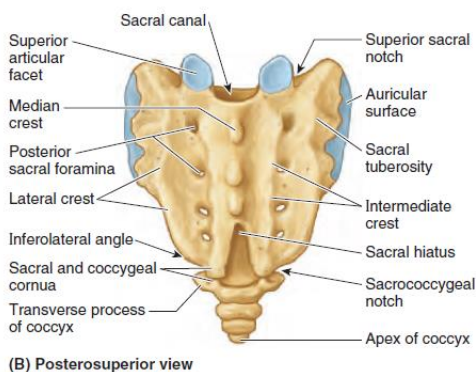
- Pelvic surface; It is concave and relatively smooth, and
- Posterior surface; has rudimentary articular processes

Coccygeal vertebra I (Co I)

- The largest and broadest of all the coccygeal vertebrae.
- Its short transverse processes are connected to the sacrum, and
- Its rudimentary articular processes form **coccygeal cornua**, which articulate with the **sacral cornua**.

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Variations in the Vertebrae

- Most people have 33 vertebrae
- But developmental errors may result in 32 or 34 vertebrae
- Estimates of the frequency of abnormal numbers of vertebrae superior to the sacrum range between 5% and 12%
- Variations in vertebrae are affected by race, sex, and developmental factors (genetic and environmental)
- An increased number of vertebrae occurs more often in males and a reduced number occurs more frequently in females

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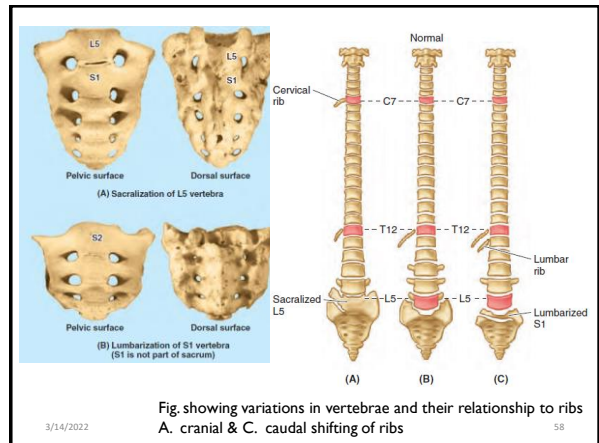
Variations in the Vertebrae

❖ Abnormal Fusion of Vertebrae

- Hemisacralization & sacralization of the L5 vertebra
 - A condition in which L5 is partly or completely incorporated into the sacrum, respectively
 - When L5 is sacralized, the L5–S1 level is strong and the L4–L5 level degenerates, often producing painful symptoms.
- Hemilumbarization & lumbarization of the S1 vertebra
 - S1 is more or less separated from the sacrum and is partly or completely fused with L5 vertebra, respectively

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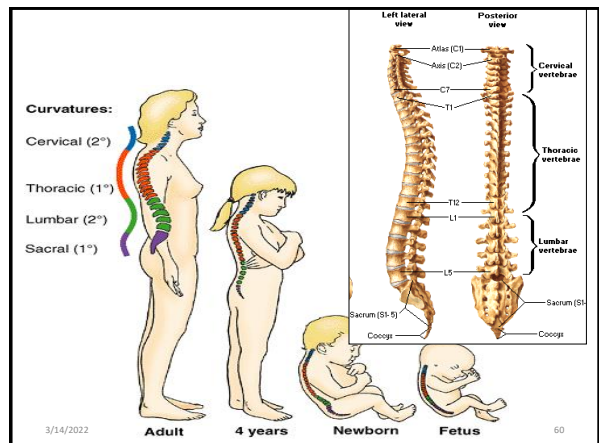
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Curvatures of the Vertebral Column

- The vertebral column in adults has four curvatures: **cervical, thoracic, lumbar, and sacral**
- 1. **Primary Curvature**
 - includes thoracic & sacral curvatures
 - **concave anteriorly**
 - Develop during the fetal period in relationship to the (flexed) fetal position
- 2. **Secondary Curvatures**
 - appear after birth, result from extension from the flexed fetal position
 - includes cervical & lumbar curvatures
 - **Concave posteriorly**

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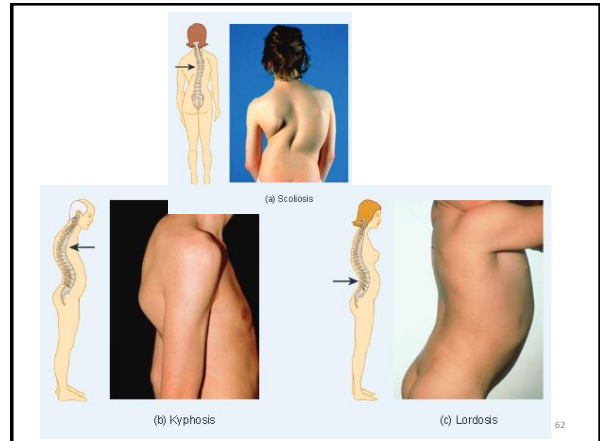
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Abnormal Curvatures of the Vertebral Column

- A. Kyphosis** (hunchback or humpback) an abnormally increased **thoracic** (forward) curvature resulting from osteoporosis;
- B. Lordosis** (swayback or saddle back)—an abnormally increased **lumbar** (backward) curvature resulting from trunk muscular weakness or osteomalacia; and
- C. Scoliosis** a condition of **lateral deviation** resulting from
 - **unequal growth of the vertebral column,**
 - pathologic erosion of vertebral bodies, or
 - asymmetric paralysis or weakness of vertebral muscles.

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Joints & ligaments of Vertebral Column

- The joints of the vertebral column include the:
 - A. Joints of the vertebral bodies
 - B. Joints of the vertebral arches
 - C. Craniovertebral (atlanto-axial and atlanto-occipital) joints
 - D. Costovertebral joints
 - E. Sacro-iliac joints

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Vertebral Ligaments

- Joints between vertebrae are reinforced and supported by numerous ligaments, which **interconnect components of the vertebral arches**.
- The well known ligaments are:
 - Anterior and posterior longitudinal ligaments,
 - Ligamentum flavum,
 - Supraspinous ligament and
 - Ligamentum nuchae
- **Other Ligaments**
 - Interspinous (between two adjacent spinous processes) &
 - Intertransverse (between two adjacent transverse processes) ligaments.

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A. Joints of the Vertebral Bodies

- The joints of the vertebral bodies are symphyses (secondary cartilaginous joints)
 - designed for weight bearing and strength
- The superior and inferior surfaces of the bodies of adjacent vertebrae are covered by **thin plates of hyaline cartilage**.
- Two adjacent vertebrae are connected by
 - **IV discs and**
 - **Ligaments** (anterior and posterior longitudinal ligaments)

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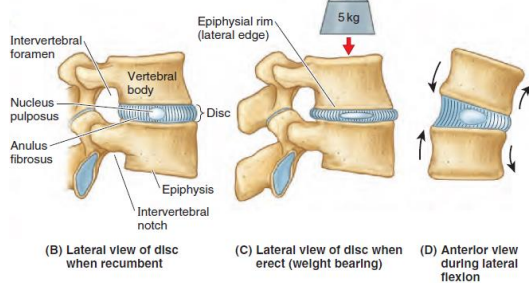
Intervertebral disc (IVD)

- A fibrocartilagenous ring that bind together the vertebral bodies.
- Extend from C2 to the sacrum (23 in number).
- There is **no** IV disc between C1 and C2 vertebrae
- Account one quarter of the length of the vertebral column below the level of C2
- Form the inferior half of the anterior border of the IV foramen
- Their main function is shock absorption
- Thickest in **the cervical and lumbar regions**.
 - where the movements of the vertebral column are greatest.

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Intervertebral disc (IVD)



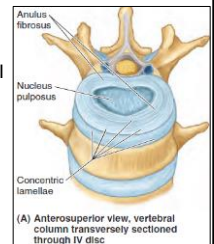
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Intervertebral disc (IVD)

Each IV disc consists of

- **anulus fibrosus**, an outer fibrous part, composed of concentric lamellae of fibrocartilage, and
- **nucleus pulposus**; a gelatinous central mass
 - ✓ Which is remnant of notochord
 - ✓ **Avascular**
 - ✓ It receives its nourishment by diffusion from blood vessels at the periphery of the **anulus fibrosus and vertebral body**



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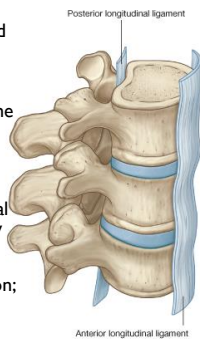
68

Anterior & posterior longitudinal ligaments

- Extend from **skull to sacrum** of vertebral column on the anterior and posterior surfaces.

Anterior longitudinal ligament;

- Attached superiorly to the base of the skull and extends inferiorly to the anterior surface of the sacrum
- is a strong, broad fibrous band that covers and connects the anterolateral aspect of the vertebral bodies and IV discs
- the only ligament that limits extension; all other IV ligaments limit forms of flexion.



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Anterior & posterior longitudinal ligaments

Posterior longitudinal ligament:

- Is weak and narrow than the anterior longitudinal ligament
- It is attached mainly to the IV discs and extend from C2 to the sacrum
- lines the anterior surface of the **vertebral canal**



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B. Joints of Vertebral Arches

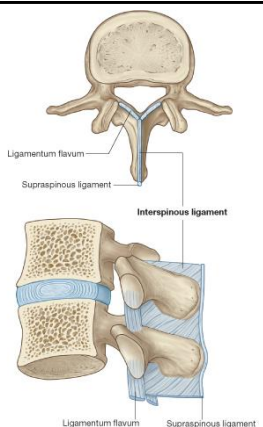
- Also called *zygapophysial joints (facet joints)*.
- Plane type of synovial joints** between the superior and inferior articular processes of adjacent vertebrae.
 - permit gliding movements between the articular processes
- Surrounded by a thin **joint capsule**
 - in the cervical region, it is thin and loose reflecting the wide range of movement.
- Accessory ligaments unite the laminae, transverse processes, and spinous processes and help stabilize the joints.

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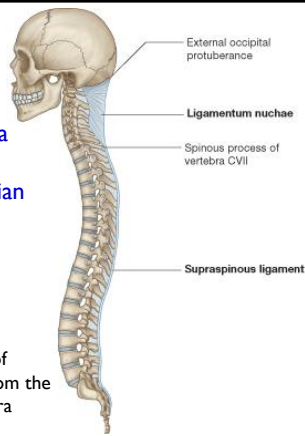
Ligaments

- Supraspinous ligament:** Connects and passes along the tips of the vertebral **spinous processes from vertebra C7 to the sacrum**
- Interspinous ligament:** connects adjacent **spines**.
- Intertransverse ligaments:** run between adjacent **transverse processes**.
- Ligamentum flavum:** connects the **laminae** of adjacent vertebrae.



Ligaments

- In cervical region
- ✓ **Ligamentum nuchae**: a triangular, sheet like structure in the median sagittal plane.
 - ✓ Its base extends the external occipital protuberance to the foramen magnum.
 - ✓ Upward continuation of **supraspinous ligament** from the spine of the C7 vertebra



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C. Craniovertebral Joints

- Synovial joints that have no IV discs
- Their design gives a wider range of movement than in the rest of the vertebral column
- The articulations involve the **occipital condyles, atlas, and axis**

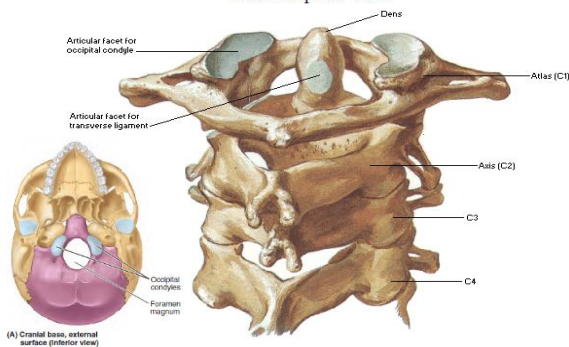
There are two sets of craniovertebral joints

- I. **Atlanto-occipital joints**; formed between atlas (C1) and occipital bone of the cranium, and
- II. **Atlantoaxial joints**; formed between atlas & axis (C2)

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Cervical Vertebrae [C1-C4], Assembled Posterosuperior View



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Atlanto-Occipital Joints

- Articulations between the superior articular surfaces of the lateral masses of the atlas and the occipital condyles
- **Condyloid** type of synovial joints.
- **Movements**
 - Flexion, extension, and lateral flexion.
 - Indicating approval (**Yes movement**)
 - **No rotation is possible.**
- **Ligaments**
 - **Anterior atlanto-occipital membrane**
 - **Posterior atlanto-occipital membrane**

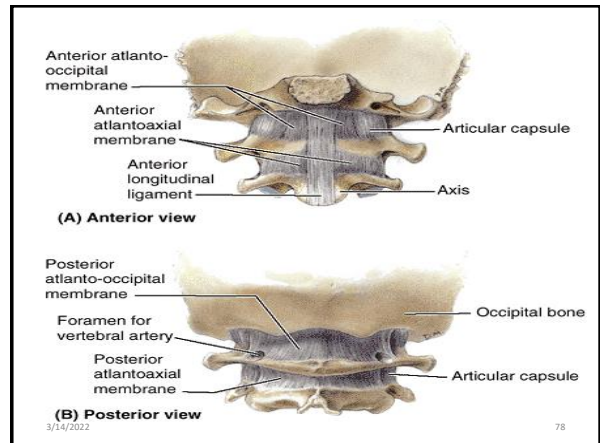
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Atlanto-Occipital Joints

Ligaments

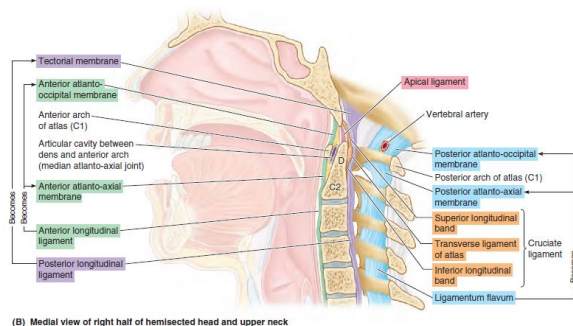
- **Anterior atlanto-occipital membrane:**
 - Connects the anterior arch of the **atlas** to the anterior margin of the foramen magnum.
 - Broader and centrally it continuous with the anterior longitudinal ligament
- **Posterior atlanto-occipital membrane:**
 - Connects the posterior arch of the atlas to the posterior margin of the foramen magnum
 - As a **ligamentum flavum** of lower vertebrae

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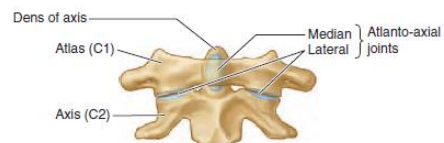
Atlanto-occipital joints ligaments



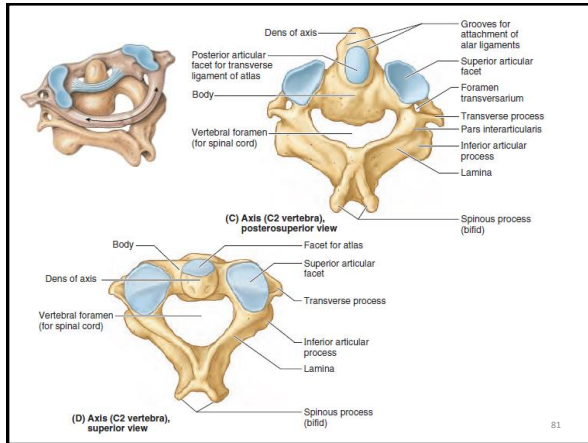
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Atlantoaxial Joints

- **Three atlantoaxial articulations:**
 - **Two (right and left) lateral atlantoaxial joints**
 - ✓ b/n the inferior facets of the lateral masses of C1 & superior facets of C2
 - ✓ are gliding-type synovial joints
 - **One median atlantoaxial joint:** is a pivot joint
 - ✓ b/n the dens of C2 and the anterior arch of the atlas



Three atlanto-axial joints



Atlantoaxial Joints

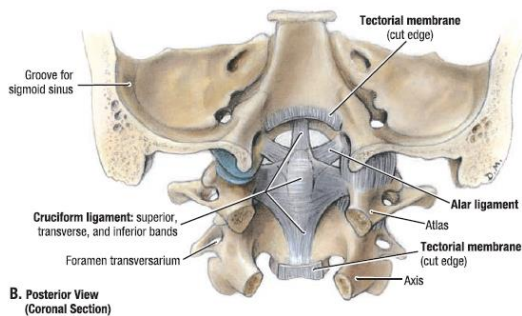
Ligaments: includes

- **Apical ligament:** median-placed structure connects the apex of the odontoid process to the anterior margin of the foramen magnum.
- **Alar ligaments:** lie one on each side of the apical ligament and connect the odontoid process to the **medial sides of the occipital condyles**.
- **Cruciate ligament:** has a transverse part and a vertical part.
 - **Transverse** part is attached on each side to the inner aspect of the lateral mass of the atlas and binds the odontoid process to the anterior arch of the atlas.

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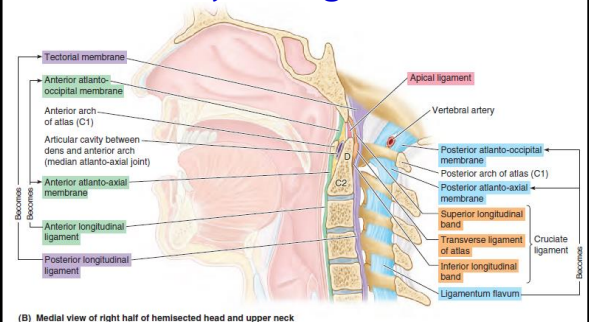
Atlantoaxial joints ligaments



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Atlantoaxial joints ligaments



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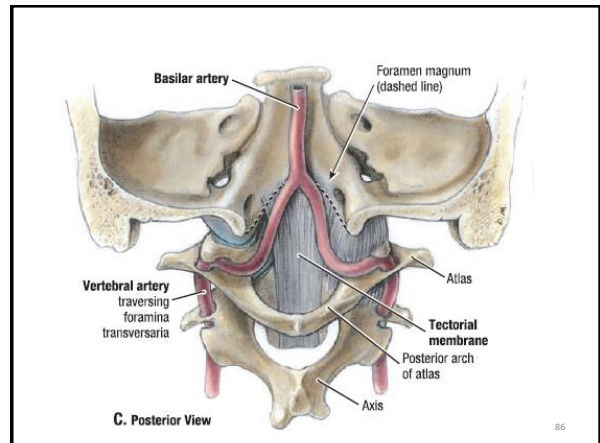
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Atlantoaxial Joints

- **Cruciate ligament...**
 - **Vertical part** runs from the posterior surface of the body of the axis to the anterior margin of the foramen magnum.
- **Membrana tectoria:** an upward continuation of the posterior longitudinal ligament.
 - It runs superiorly from the body of C2 through the foramen magnum to attach to basilar part of occipital bone.
 - **Covers** the posterior surface of the odontoid process and the above three ligaments.

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Atlantoaxial Joints

Movement at all three atlantoaxial joints

- permits the head to be turned from side to side,
- as occurs when rotating the head to indicate disapproval (**the no movement**)

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Nerve Supply of Vertebral Joints

Joints of the Vertebral Bodies

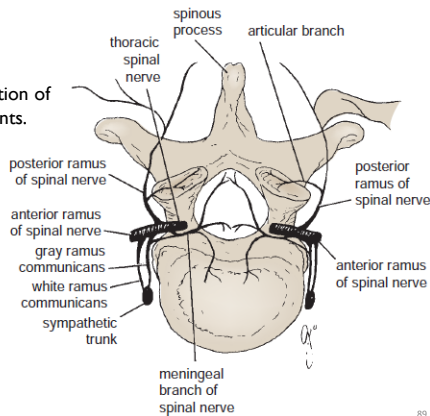
- Innervated by the small meningeal branches of each spinal nerve
- It arises from the spinal nerve as it exits from the intervertebral foramen.
- re-enters the vertebral canal and supplies the **meninges, ligaments, and intervertebral discs.**
- At any particular level joints receive nerve fibers from two adjacent spinal nerves.

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Fig.

- The innervation of vertebral joints.



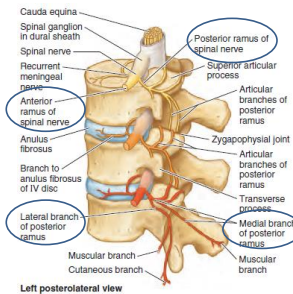
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Joints of Vertebral Arches

Innervation:

- Posterior rami divide into
 - ✓ Medial and
 - ✓ Lateral branches.
- The medial branch gives rise to articular branches that innervate the zygapophysial joint at that level, and to the joint one level inferior to its exit.
 - therefore, each joint is supplied by two nerves.



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Movements of the Vertebral Column

- The range of movement of the vertebral column varies according to the region and the individual
- The vertebral column is capable of
 - **Flexion & Extension;** occurring primarily in the cervical and lumbar regions
 - **Lateral flexion and lateral extension;** occurring mostly in the cervical and lumbar regions
 - **Rotation;** occurs primarily at the craniovertebral joints and thoracic region

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Movements of the Vertebral Column

- Movements of the vertebral column are more free in the **cervical** and **lumbar** regions than elsewhere
- ❖ **Note:** Vertebral column in thoracic region is relative stable because
 - It's IV discs are thin relative to the size of the vertebral bodies
 - It connected to the sternum by the ribs and costal cartilages.

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Movements of the Vertebral Column

❖ The range of vertebral column movement is limited by:

- Thickness, elasticity and compressibility of IVDs
- Shape and orientation of zygapophysial joints
- Tension of the joint capsules of zygapophysial joints
- Resistance of the back muscles and ligaments
→ e.g., ligamenta flava and post. longitudinal ligament
- Attachment to the thoracic (rib) cage
- Bulk of surrounding tissue

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Movements of the Vertebral Column

❖ **Flexion, extension, lateral flexion, and rotation of the neck are especially free because the:**

- IV discs are thick relative to the size of the vertebral bodies at this level
- Surfaces of zygapophysial facet are relatively large and the joint planes are almost horizontal
- Joint capsules of the zygapophysial joints are loose
- Neck is relatively slender (less surrounding soft tissue).

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Movements of the Vertebral Column

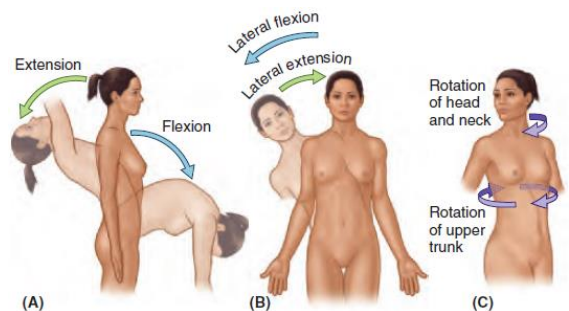
- The sagittally oriented joint planes of the lumbar region are conducive to flexion and extension
➢ however, the interlocking articular processes here prevent rotation
- **Lateral flexion** of the vertebral column is greatest in the cervical and lumbar regions
- In the thoracic region the joint planes lie on an arc that is centered on the vertebral body, permitting rotation

→ But flexion and lateral flexion are limited

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Movements of the Vertebral Column



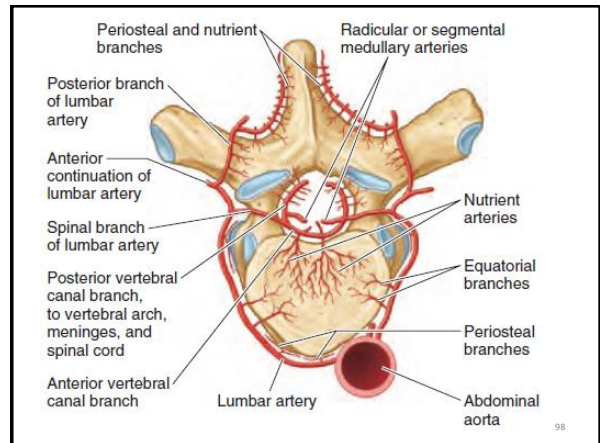
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Vasculature of the Vertebral Column

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Vasculature of the Vertebral Column

- ❖ **Vertebrae are supplied by**
 - Periosteal, equatorial and spinal branches of
 - the major cervical arteries and
 - *Segmental arteries* of the trunk and
 - These branches occur at all levels of the vertebral column, and from the following arteries:

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Vasculature of the Vertebral Column

Parent arteries of periosteal, equatorial, and spinal branches include

Neck region	Thoracic region	Abdominal region	Pelvic region
▪ Vertebral ▪ Ascending cervical	▪ Posterior intercostal	▪ Subcostal ▪ Lumbar	▪ Iliolumbar ▪ Lateral sacral ▪ Medial sacral

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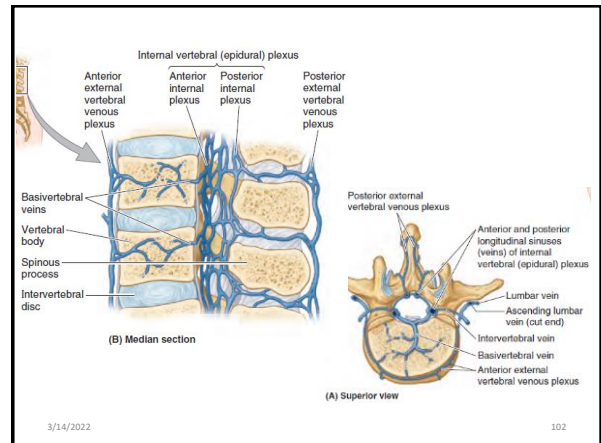
Vasculature of the Vertebral Column

Venous drainage

- ❖ Spinal veins form venous plexuses;
 - **Internal vertebral venous plexus** (epidural venous plexuses); inside the vertebral canal
 - **External vertebral venous plexuses**, outside the vertebral canal
- These plexuses communicate through the **intervertebral foramina**
- Drain into the **vertebral veins of the neck** and **segmental (intercostal, lumbar, and sacral) veins** of the trunk

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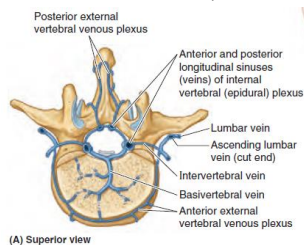
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Intervertebral veins

Con't

- Receive veins from the spinal cord and vertebral venous plexuses pass through IV foramina to
- Drain into the **vertebral veins** of the neck and **segmental (intercostal, lumbar, and sacral) veins** of the trunk

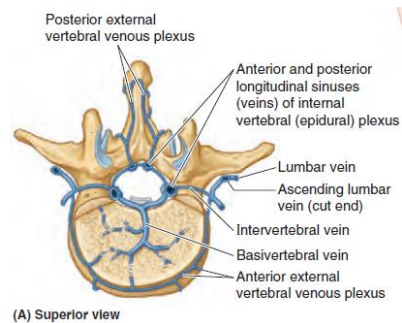


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(A) Superior view

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Vasculature of the Vertebral Column



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(A) Superior view

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Nerves of the Vertebral Column

- ❖ Zygapophysial joints
 - Innervated by articular branches of the medial branches of the posterior rami
- ❖ Other part of vertebral column
 - Innervated by **recurrent meningeal** branches of the spinal nerves
 - arise from the mixed spinal nerve, or from the anterior ramus.

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Nerves of the Vertebral Column

Recurrent meningeal of spinal nerve give branches **outside and inside** the vertebral canal

- ❖ **Outside the canal**
 - anterolateral aspect of the vertebral bodies and IV discs.
 - periosteum and especially the anuli fibrosi and anterior longitudinal ligament.

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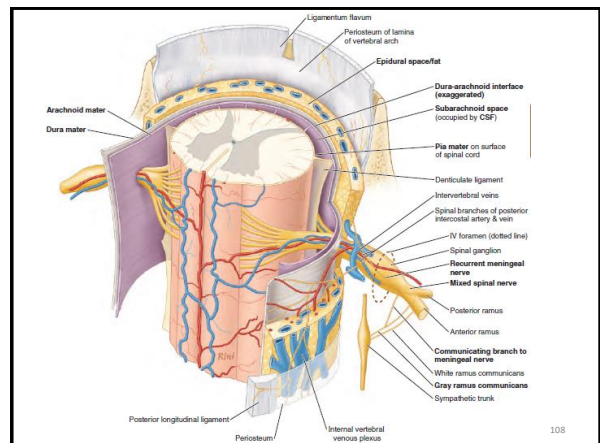
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Nerves of the Vertebral Column

- ❖ **Inside the canal,**
 - transverse, ascending, and descending branches distribute nerve fibers to the:
 - Periosteum (covering the surface of the posterior vertebral bodies, pedicles, and laminae).
 - Ligamenta flava.
 - Anuli fibrosi of the posterior and posterolateral aspect of the IV discs.
 - Posterior longitudinal ligament.
 - Spinal dura mater.
 - Blood vessels within the vertebral canal.

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Muscles of the Back

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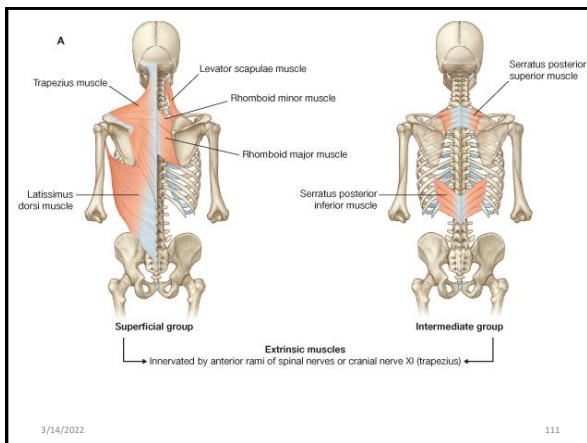
Muscles of the Back

Two major groups of muscles in the back

- **The extrinsic back muscles** include
 - **Superficial muscles:** produce and control limb movements
 - **Intermediate muscles:** produce and control respiratory movements.
- **The intrinsic (deep) back muscles**
 - Specifically act on the vertebral column, producing its movements and maintaining posture

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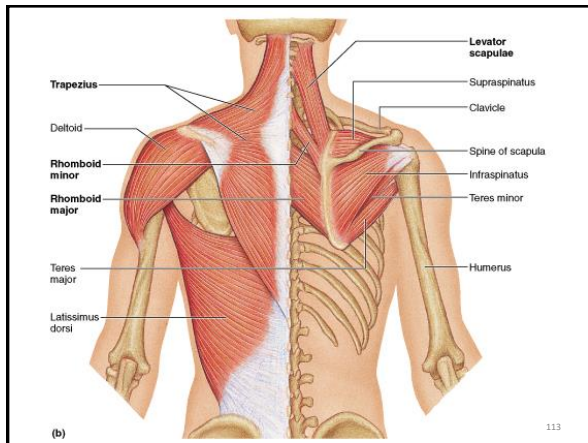
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Superficial extrinsic back muscles

- are posterior axio-appendicular muscles
- Connect the pectoral girdle and humerus to the trunk and produce and control limb movements
- **Includes:** trapezius, latissimus dorsi, levator scapulae, and rhomboids mm
- Most part these muscles receive their nerve supply from the anterior rami of cervical nerves and act on the upper limb
- The trapezius receives its motor fibers from a cranial nerve, the spinal accessory nerve (CN XI)

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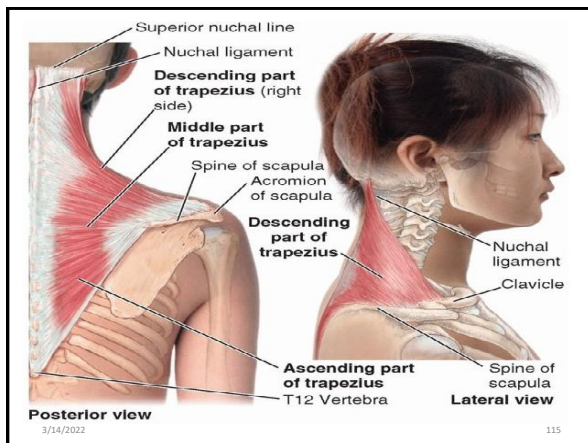


Trapezius

- Large, triangular muscle covering the posterior aspect of the neck and the superior half of the trunk
- The two sides of the muscle form a trapezium
- Attaches the pectoral girdle to the cranium and vertebral column
- Its fibers are divided into three parts:
 1. Superior fibers- elevate the scapula
 2. Middle fibers-retract the scapula
 3. Inferior fibers- depress the scapula

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Trapezius

- **Origin:** -Medial third of superior nuchal line
 - External occipital protuberance
 - Nuchal ligament & Spinous processes of C7 - T12 vertebrae
- **Insertion:** Lateral third of clavicle; acromion and spine of scapula
- **Nerve Supply:** Accessory nerve (CN XI)
- **Main Action:**
 - ✓ Descending part elevates;
 - ✓ Ascending part depresses; and
 - ✓ Middle part (or all parts together) retracts scapula
 - ✓ Descending and ascending parts act together to rotate glenoid cavity superiorly

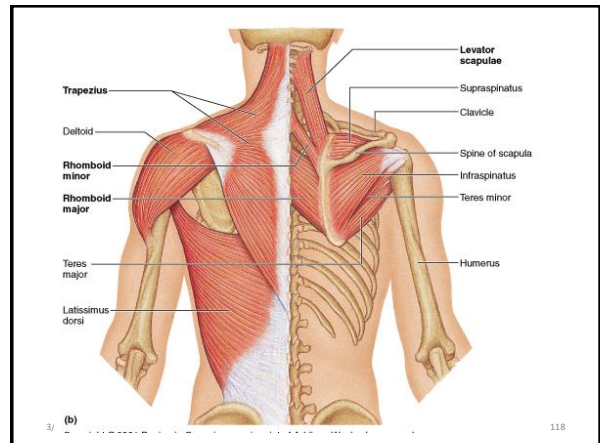
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Latissimus dorsi

- Latissimus dorsi (L. widest of the back)
- Large, fan-shaped muscle passing from the trunk to the humerus
- In combination with the pectoralis major:
 - is a powerful adductor of the humerus
 - is also useful in climbing a tree, as it raises the trunk to the arm (thus Climbing muscle)

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Latissimus dorsi

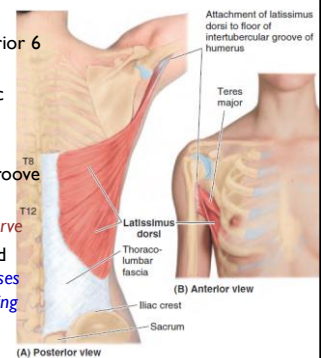
- **Origin**
 - Spinous processes of inferior 6 thoracic vertebrae,
 - Thoracolumbar fascia, Iliac crest, and
 - Inferior 3 or 4 ribs
- **Insertion:** intertubercular groove of humerus
- **Innervation:** Thoracodorsal nerve
- **Action:** Extends, adducts, and medially rotates humerus; *raises body toward arms during climbing*

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Latissimus dorsi

- **Origin**
 - Spinous processes of inferior 6 thoracic vertebrae,
 - Thoracolumbar fascia, Iliac crest, and
 - Inferior 3 or 4 ribs
- **Insertion:** intertubercular groove of humerus
- **Innervation:** Thoracodorsal nerve
- **Action:** Extends, adducts, and medially rotates humerus; *raises body toward arms during climbing*



(A) Posterior view

(B) Anterior view

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Levator scapulae

- Lies deep to the SCM & Trapezius
- True to its name, it acts to elevate the scapula
- **Origin:** Posterior tubercles of transverse processes of C1 - C4 vertebrae
- **Insertion:** Medial border of scapula superior to spine
- **Nerve Supply:** **Dorsal scapular** (C4 & C5) and **cervical** (C3, C4) nerves
- **Main Action:**
 - Elevates scapula and tilts its glenoid cavity inferiorly by rotating scapula

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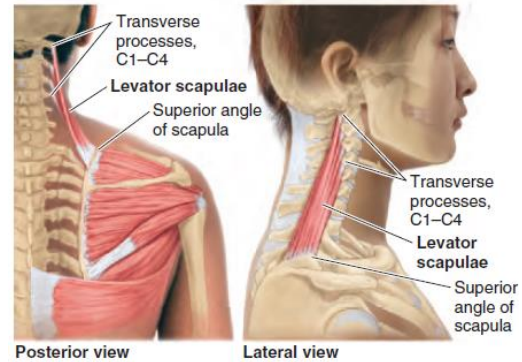


FIGURE 6.27. Levator scapulae.

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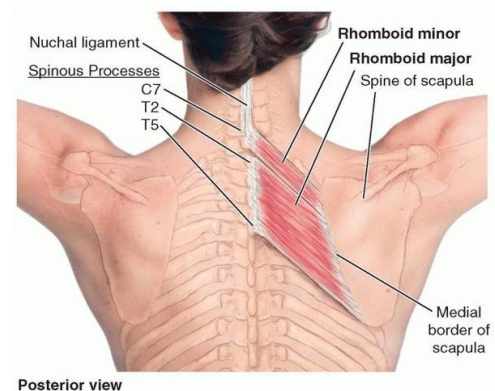
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Rhomboids (major and minor)

- Have a rhomboid appearance
- They lie deep to the trapezius and form broad parallel bands that pass inferolaterally from the vertebrae to the medial border of the scapula
- **Origin:**
 - Minor: nuchal ligament; spinous processes of C7 and T1 vertebrae
 - Major: spinous processes of T2 to T5 vertebrae
- **Insertion:**
 - Minor: medial end of scapular spine
 - Major: medial border of scapula from level of spine to inferior angle

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Posterior view

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Rhomboids (major and minor)

- **Rhomboid Major:** **THINNER**, ~2x wider than **rhomboid minor**
- **Rhomboid minor-THICKER**, superior in position
- **Innervation:**
 - Dorsal scapular nerve
- **Action:**
 - Retract scapula and rotate it to depress glenoid cavity
 - fix scapula to thoracic wall during movements of upper limb

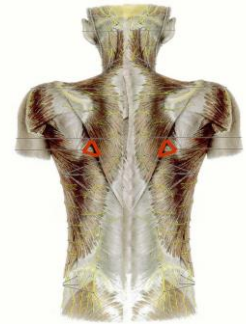
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Muscular Triangles of the Back

Auscultatory Triangle

- The site on the back where breath sounds may be most easily heard with a **stethoscope**.
- Formed by the
 - ✓ **Inferior:** Superior border of latissimus dorsi,
 - ✓ **Lateral:** Medial border of scapula, and
 - ✓ **Medial:** Inferolateral border of trapezius
 - ✓ **Floor:** Rhomboid major



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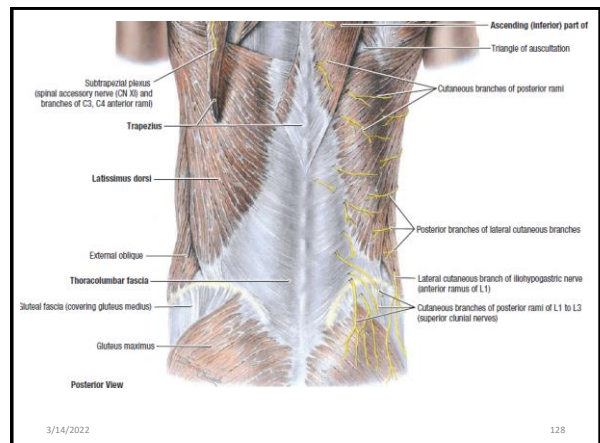
Muscular Triangles of the Back

Lumbar Triangle

- It may be the site of
 - ✓ an abdominal hernia and
 - ✓ pus may emerge from the abdominal wall.
- **Boundaries**
 - ✓ **Posterior:** latissimus dorsi,
 - ✓ **Anterior:** Posterior border of the external oblique muscle of the abdomen, and
 - ✓ **Inferior:** Iliac crest
 - ✓ **Floor:** Internal oblique muscle

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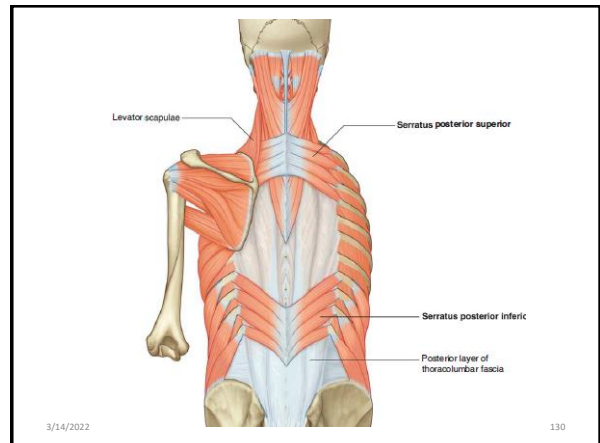
Intermediate extrinsic back muscles

Serratus posterior muscles;

- are thin muscles, commonly designated as superficial respiratory muscles.
- Serratus posterior superior** lies deep to the rhomboid muscles, and
- Serratus posterior inferior** lies deep to the latissimus dorsi.
- Both are innervated by **intercostal nerves**

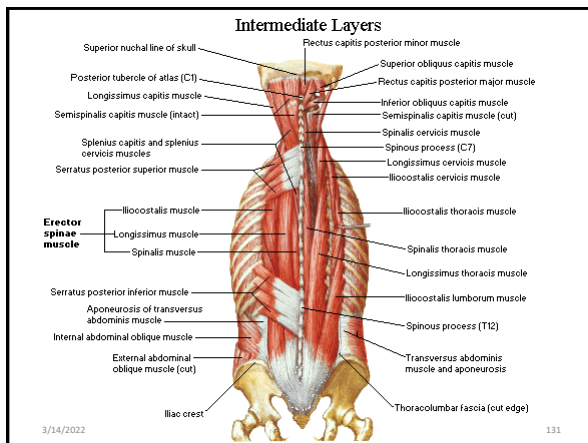
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Intermediate extrinsic back muscles

- Serratus posterior superior**
 - Origin:** Nuchal ligament, spinous processes of C7 to T3 vertebrae
 - Insertion:** Superior borders of 2nd to 4th ribs
 - Action:** elevate ribs
 - Innervation:** 2nd to 5th intercostal nerves
- Serratus posterior inferior**
 - Origin:** Spinous processes of T11 to L2 vertebrae
 - Insertion:** Inferior borders of 8th to 12th ribs near their angles
 - Action:** depress ribs
 - Innervation:** T9 to T12 thoracic spinal nerves

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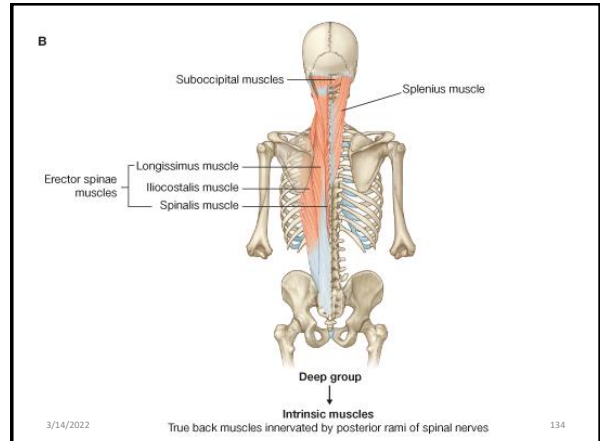
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Intrinsic Back Muscles

- Also called muscles of back proper, deep back muscles
- Because:
 - Act to maintain posture and control movements of the vertebral column &
 - Innervated by the posterior rami of spinal nerves
- These muscles extend from the **pelvis to the cranium** & enclosed by **deep fascia**

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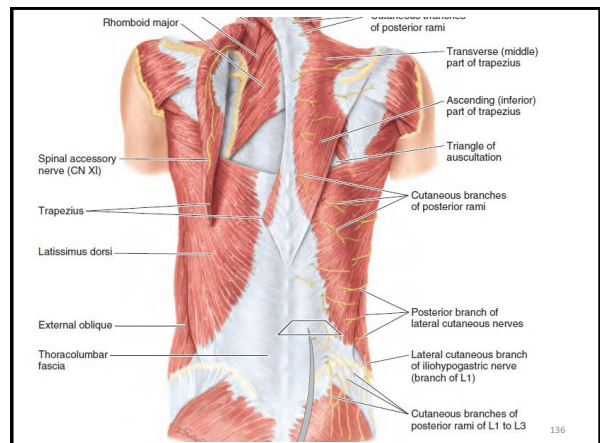
Intrinsic Back Muscles

Deep fascia

- Medially;** attaches to the
 - nuchal ligament,
 - tips of the spinous processes of the vertebrae,
 - supraspinous ligament, and
 - median crest of the sacrum
- Laterally;** attaches to the
 - cervical and lumbar transverse processes and to the angles of the ribs
- In thoracic and lumbar regions it is called **thoracolumbar fascia**

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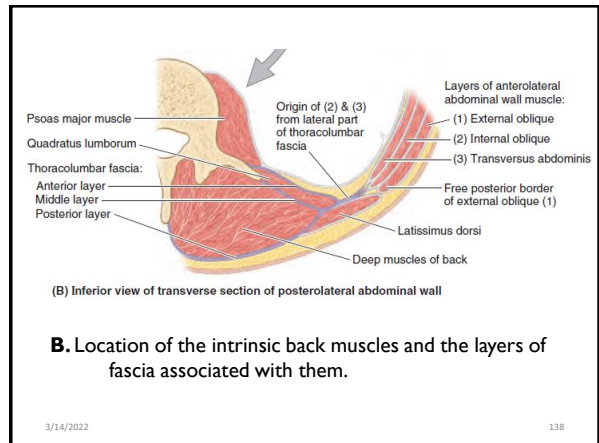
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Thoracolumbar fascia

- Invests the deep muscles of the back,
- **Have**
- I. **Anterior layer:** lies anterior to the quadratus lumborum and attaches to the vertebral **transverse process**, and
- II. **Middle layer:** lies b/n quadratus lumborum and erector spinae attaches to the vertebral **transverse process**
- III. **Posterior layer** that lies posterior to the erector spinae and attaches to the **spinous processes**.
- Provides the **origins** for the latissimus dorsi and the internal oblique and transversus abdominis muscles.

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Intrinsic Back Muscles

- According to their relationship to the surface, intrinsic (deep) back muscles are grouped in to three layers
 - Superficial,
 - Intermediate, and
 - Deep layers

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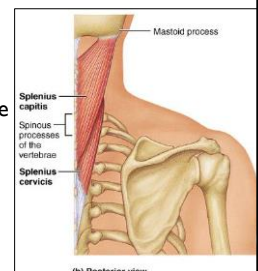
Superficial Layer of Intrinsic Back Muscles

(Spinotransverse Group)

Splenius muscles (L. musculi splenii)

- lie on the lateral and posterior aspects of the neck, covering the vertical muscles like a bandage
- Cover & hold the deep neck muscles in position.
- Consisting of
 - Splenius cervicis
 - Splenius capitis

→ Arise from the **midline** and **extend superolaterally** to the cervical vertebrae and cranium respectively.



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Superficial Layer of Intrinsic Back Muscles

Splenius muscles

- **Origin:** Nuchal ligament and spinous processes of C7–T6 vertebrae
- **Insertion**
 - ***Splenius capitis*:** fibers run superolaterally to mastoid process of temporal bone and lateral third of superior nuchal line of occipital bone
 - ***Splenius cervicis*:** tubercles of transverse processes of C1–C3 or C4 vertebrae
- **Innervation:** Posterior rami of spinal nerves

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Superficial Layer of Intrinsic Back Muscles

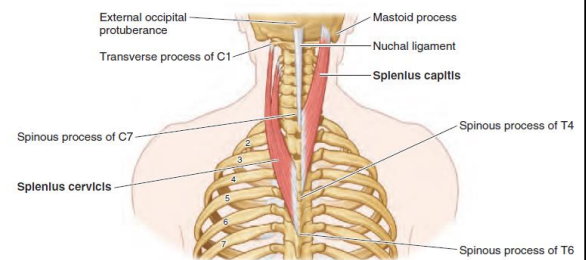


FIGURE 4.30. Superficial layer of intrinsic back muscles (splenius muscles).

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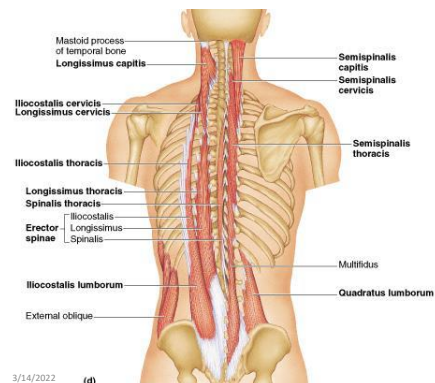
Intermediate Layer of Intrinsic Back Muscles

Grouped as **erector spinae** muscles (Sacrospinalis Group)

- Lie in a groove on each side of the vertebral column between the **spinous processes** centrally and the **angles of the ribs** laterally
- **Chief extensor** of the vertebral column
- **Divided** into three columns:
 - Lateral → **Iliocostalis**
 - Intermediate → **Longissimus** and
 - Medial → **Spinalis**
- Each column is divided regionally into three parts according to the superior attachments

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Intermediate of Intrinsic Back Muscles



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(d)

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Intermediate Layer of Intrinsic Back Muscles

- A. Iliocostalis** forms the lateral column,
 - Iliocostalis lumborum
 - Iliocostalis thoracis
 - Iliocostalis cervicis
- B. Longissimus** forms the intermediate column, and
 - Longissimus thoracis
 - Longissimus cervicis
 - Longissimus capitis
- C. Spinalis** forms the medial column.
 - Spinalis thoracis
 - Spinalis cervicis
 - Spinalis capitis

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Erector spinae (Intermediate intrinsic back muscles)

- **Origin**
 - Iliac crest, posterior surface of sacrum, sacro-iliac ligaments,
 - Sacral and inferior lumbar spinous processes, and supraspinous ligament
- **Insertion**
 - **Iliocostalis:** lumborum, thoracis, cervicis;
→ fibers run superiorly to angles of lower ribs and cervical transverse processes

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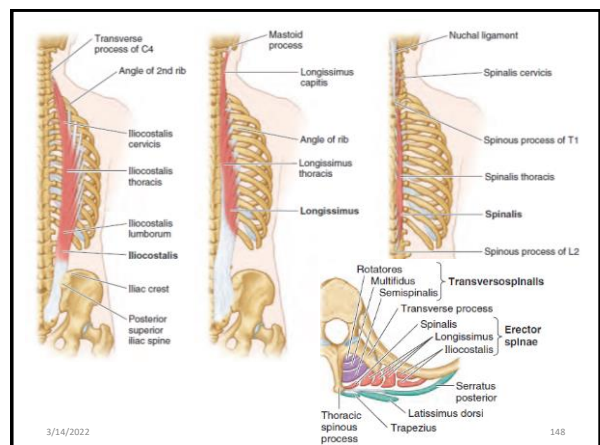
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Erector spinae (Intermediate intrinsic back muscles)

- **Insertion**
 - **Longissimus:** thoracis, cervicis, capitis;
→ Fibers run superiorly to ribs between tubercles and angles
→ to transverse processes in thoracic and cervical regions, and
→ to mastoid process of temporal bone
 - **Spinalis:** thoracis, cervicis, capitis;
→ Fibers run superiorly to spinous processes in the upper thoracic region and to cranium
- **Innervation:** Posterior rami of spinal nerves

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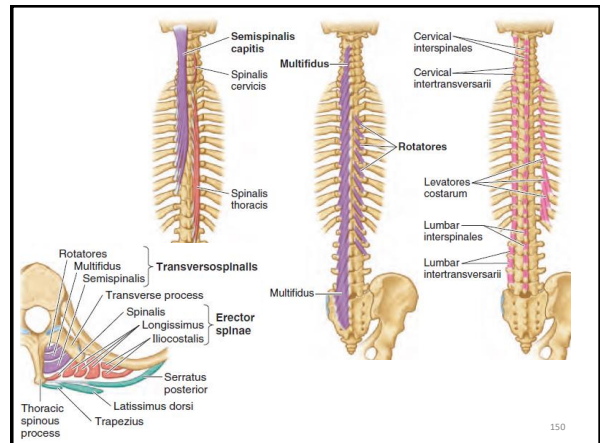
Deep layer of intrinsic back muscles

Transversospinalis group

- Deep to the erector spinae & an obliquely disposed group of much shorter muscles
- Occupy the "gutter" b/n the transverse & the spinous processes.
- Consisting of
 - Semispinalis,
 - Multifidus, and
 - Rotatores
- Originate from transverse processes of vertebrae and pass to spinous processes of more superior vertebrae.

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Deep layer of intrinsic back muscles (Transversospinalis)

Semispinalis

- Superficial member of the group.
- It is divided into three parts according to the superior attachments:
 - Semispinalis capitis,
 - Semispinalis thoracis, and
 - Semispinalis cervicis
- Fibers arise from transverse processes of C4–T12 vertebrae to occipital bone and spinous processes in thoracic and cervical regions

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Deep layer of intrinsic back muscles (Transversospinalis)

Multifidus

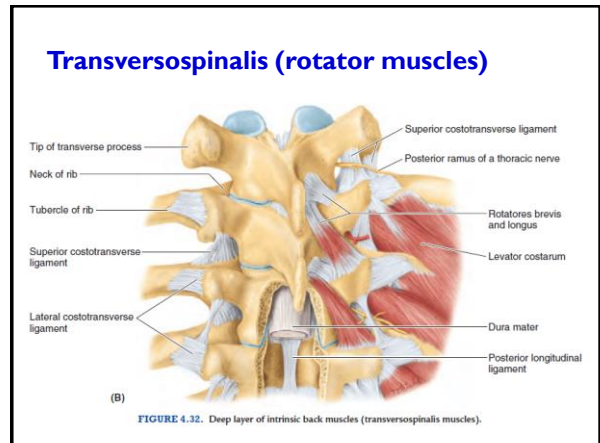
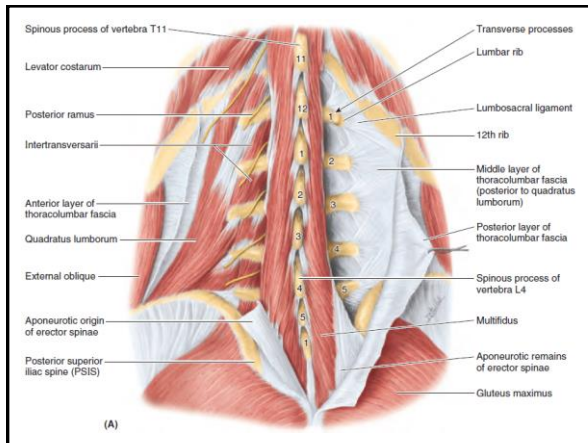
- Middle layer of the group
- Consists of short, triangular muscular bundles that are thickest in the lumbar region.

Rotatores or rotator muscles

- Deepest of the three layers of transversospinal muscles
- Best developed in the thoracic region.
- Fibers run from transverse processes of vertebrae to junction of lamina and transverse process or spinous process of vertebra

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Intrinsic Back Muscles

IV. Other segmental deep back muscles

- Are innervated by the dorsal primary rami of the spinal nerves.
- Consist of the following
 - A. Interspinales:** Run between adjacent spinous processes and aid in extension of the vertebral column.
 - B. Intertransversarii:** Run between adjacent transverse processes and aid in lateral flexion of the vertebral column
 - C. levatores costarum:** Extend from the transverse processes to ribs, elevate the ribs, and are innervated by the intercostal nerves.

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Suboccipital & Deep Neck Muscles

Suboccipital region

- is a muscle "compartment" deep to the superior part of the posterior cervical region,
- underlying the trapezius, sternocleidomastoid, splenius, and semispinalis muscles.
- It is a pyramidal space inferior to the external occipital prominence of the head that includes the posterior aspects of vertebrae C1 and C2.
- Consist of four small muscles;
 - Two rectus capitis posterior (major and minor) and
 - Superior and inferior obliquus capitis muscles.

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Suboccipital M	Origin	Insertion
Rectus capitis posterior major	Spinous process of vertebra C2	Lateral part of inferior nuchal line of occipital bone
Rectus capitis posterior minor	Posterior tubercle of posterior arch of vertebra C1	Medial part of inferior nuchal line of occipital bone
Obliquus capitis inferior	Post tubercle of post arch of C2	Transverse process of vertebra C1 (atlas)
Obliquus capitis superior	Transverse process of vertebra C1	Occipital bone b/n superior and inferior nuchal lines

Obliquus capitis inferior is the only "capitis" muscle that has no attachment to the cranium

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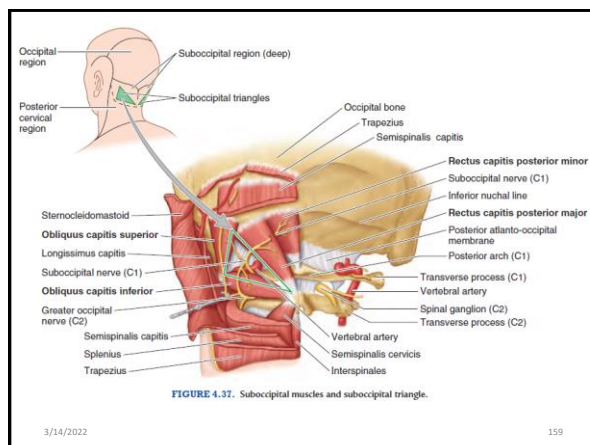
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Suboccipital & Deep Neck Muscles

- All four muscles are innervated by **suboccipital nerve** (*posterior ramus of C1*)
- The nerve emerges within the **suboccipital triangle**.
 - b/n vertebral artery above and posterior arch of atlas below

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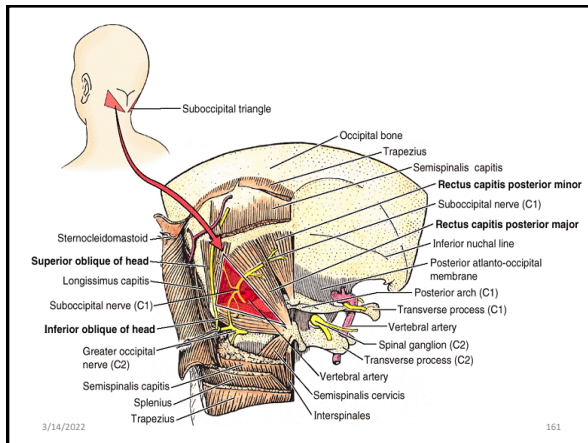
Suboccipital & Deep Neck Muscles

Suboccipital triangle

- Boundaries
 - Medially: rectus capitis posterior major muscle,
 - Laterally: obliquus capitis superior muscle, and
 - Inferiorly: obliquus capitis inferior muscle.
- Roof: semispinalis capitis and longissimus capitis.
- Floor: posterior arch of the atlas and posterior atlanto-occipital membrane.
- Contains:** vertebral artery and suboccipital nerve & vessels.

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Clinical Correlation of Vertebral Column

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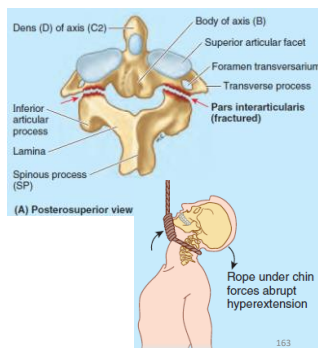
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Fracture and Dislocation of C2

- Fractures of the **vertebral arch of the axis** are one of the most common injuries of the cervical Vertebrae

Hangman's fracture:

- Fracture at the **pars interarticularis** of the axis (C2),
- Due to **hyperextension of the head over the neck**.
- In this fracture, the **cruciform ligament** is torn and the **spinal cord** is crushed, causing death.



Fracture and Dislocation of C2

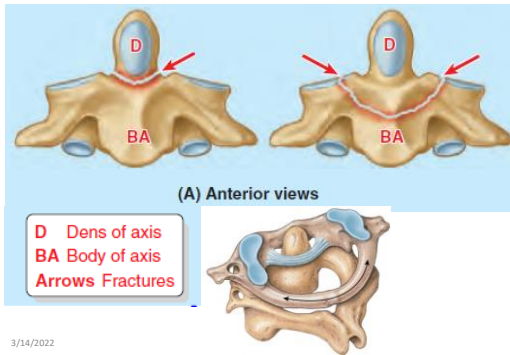
Fractures of the dens are

- The transverse ligament of atlas is stronger than dens of the C2.
- Fractures of the dens make up about 40% of fractures of the axis.
- Commonly dens fracture occurs at
 - Its base:** do not reunite because
 - the transverse ligament of the atlas becomes interposed between fragments and
 - the separated fragment (the dens) no longer has a blood supply.
 - Vertebral body** inferior to the base of the dens.
 - heals more readily because the fragments retain their blood supply.
- Other dens fractures result from abnormal ossification patterns.

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Fractures of the dens at



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Cervical Ribs

- A *cervical rib* is a relatively common anomaly.
- Supernumerary (extra) rib: place pressure on structures that emerge from the **superior thoracic aperture**,
 - notably the subclavian artery or inferior trunk of the brachial plexus, and
 - may cause thoracic outlet syndrome.

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Anomalies of the Vertebrae

Spina bifida

- Defective closure of the vertebral arch associated with maternal folic acid deficiency
- Usually occurs in the vertebral arch of L5 and/or S1
- It may be classified as
 - I. Spina bifida occulta
 - II. Spina bifida cystica

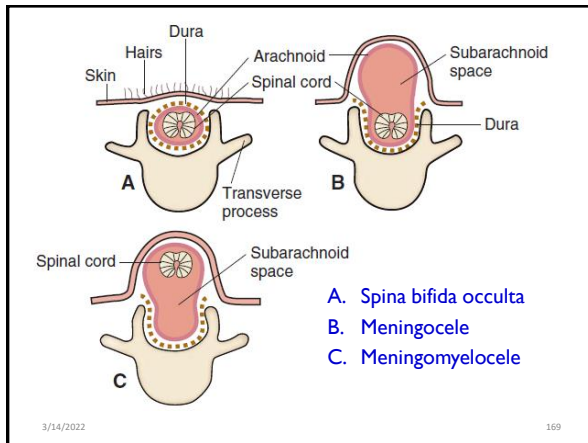
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Anomalies of the Vertebrae

Spina bifida occulta

- A common congenital anomaly of the vertebral column
- The laminae (neural arches) of L5 and/or S1 fail to fuse posterior to the vertebral canal
- The defect is concealed by the overlying skin
→ but its location is often indicated by **a tuft of hair**

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Anomalies of the Vertebrae

Spina bifida cystica

- Severe types of spina bifida, one or more vertebral arches may fail to develop completely
- Associated with herniation of
 - I. Meninges called **meningocele**, and
 - II. Meninges and/or spinal cord called **meningomyelocele**



Infant with spina bifida cystica

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Anomalies of the Vertebrae

II. Meningomyelocele

- Neurological symptoms are usually present
e.g., paralysis of the limbs and
disturbances in bladder and bowel control
- Result from **neural tube defects**, such as the defective closure of the neural tube during the 4th week of embryonic development
- It is associated with maternal folic acid deficiency

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Herniation of the Nucleus Pulposus

- Herniation or protrusion of the gelatinous nucleus pulposus into or through the anulus fibrosus is a well-recognized cause of low back and lower limb pain
- It commonly occurs **posterolaterally**,
 - ✓ where the anulus fibrosus is relatively thin and
 - ✓ does not receive support from either the posterior or the anterior longitudinal ligaments

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Herniation of the Nucleus Pulposus

- **Posterolateral herniations** of the nucleus pulposus are most common in the lumbar and lumbosacral regions
 - ✓ Because the IV discs are largest in these regions and movements are consequently greater;
- Approximately 95% of lumbar disc protrusions occur at the L4-L5 or L5-S1 levels

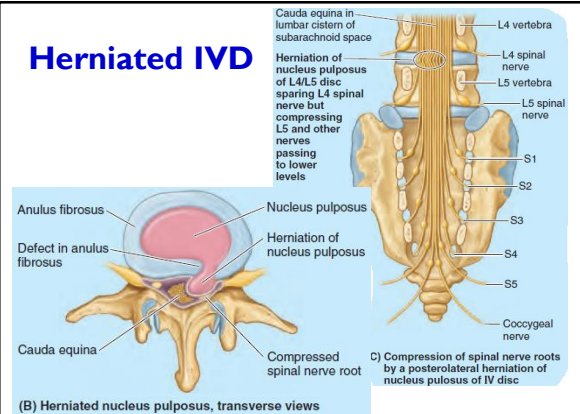
Note:

- The general rule is that when an IV disc protrudes, it usually compresses the nerve root numbered one inferior to the herniated disc;
 - e.g. An L4-L5 IV disc herniation would likely compress the L5 nerve (not L4 nerve roots)

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Herniated IVD



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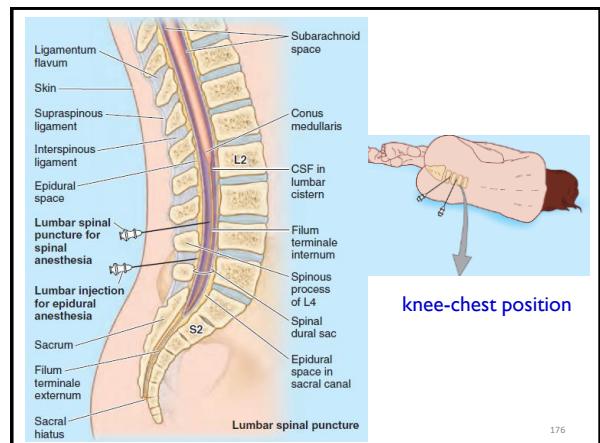
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Lumbar Spinal Puncture

- Is the tapping of the subarachnoid space in the lumbar region (lumbar cistern),
- Usually between the laminae of vertebrae L3 and L4 or vertebrae L4 and L5.
- LP is performed with the patient lying on the side with the back and hips flexed (**knee-chest position**)
- It allows withdrawal of a sample of CSF for microbial or chemical analysis and also allows introduction of anesthesia, drugs, or radiopaque material into the subarachnoid space.

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Thank you

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